LPS Local Dimension Regularization: Implementation Notes

Precise test and experiment specifications, and a VALENCIA-inspired non-manifold example set with known ground-truth dimension

Internal implementation note — LPS / `geosmooth` program

source: `~/current_projects/geosmooth/dev/methods/lps/specs/lps_local_dimension_regularization_implementation_notes.`

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Abstract

This is the implementation companion to the expository note *Local Dimension Regularization in Local Polynomial Smoothing* and to the *LPS Experimental Plan*. It specifies, precisely enough to implement without further design, two distinct kinds of evaluation. **Tests** are deterministic, pass/fail checks of the local-dimension-field machinery, run on *synthetic* data whose ground-truth dimension field we designed and therefore know exactly. **Experiments** are exploratory studies of the *real* VALENCIA 16S data (the France et al. collection, $n \approx 13{,}231$), restricted to relative abundances on the simplex; they have no ground truth, and their purpose is to characterize the data and to *guide next steps*. The two are connected by a single loop: the experiments measure the structure of the real data — which support patterns (naive strata) occur, and how the mass concentrates near lower-dimensional faces — and those measurements parameterize a family of *purely synthetic* generators that reproduce that structure while carrying a known local-dimension field. Those generators become the expanded non-manifold example set on which the tests run. Every test and experiment is stated with its objective, exact procedure or configuration, the statistic computed, and a numeric acceptance criterion or decision rule.

On the figures. All figures are original schematics drawn for this note; the "VALENCIA" panels are illustrative sketches of the *shape* of the outputs the experiments will produce, not computed results (the data are not used in producing this document).

Contents

1 Scope, and the tests–experiments distinction

The expository note argued, conceptually, that the local chart dimension in LPS should be estimated as a *field* over the neighbor graph and regularized with an ℓ_1 (total-variation) penalty so that it denoises within strata while keeping the jumps between strata sharp. This note turns that argument into an implementable validation program. The single most important design decision is to keep two kinds of evaluation rigorously separate.

- **Tests** (Section 3) are deterministic and pass/fail. They run on synthetic data whose true local-dimension field is *known by construction*, so "correct" is well defined and a continuous-integration suite can enforce it. They answer: *does the estimator do what it claims?*
- **Experiments** (Section 4) are exploratory. They run on the real VALENCIA data, which has *no* ground-truth dimension, so they produce descriptive statistics and figures rather than verdicts. They answer: *what does the real data look like, and what should we build and try next?*

These must not be conflated. A statistic measured on real data (an estimated effective dimension, say) is evidence about the data and about our estimator's behavior, but it can never be a pass/fail test, because there is nothing to check it against. Conversely, a test that passed on a synthetic toy says nothing about realism. The two are joined by the loop in Figure 1: the experiments characterize the real data; that characterization parameterizes synthetic generators (Section 5) that carry a known dimension field; and the tests run on those generators. Real data supplies *realism*; synthetic data supplies *ground truth*; neither alone suffices.

> **In plain terms** We are checking a tool that guesses the "local dimension" of a data cloud. To check a tool you need an answer key, and real biological data does not come with one. So we do two things. We *explore* the real microbiome data to learn its shape (no answer key — this guides what to build). Then we *build* fake data that has the same shape but where we wrote the answer key ourselves, and we grade the tool against that. The fake data is realistic because the exploration told us how; it is gradeable because we made it.

How tests and experiments fit together: real data is explored (no truth) to build synthetic generators (known truth) that the tests run on.

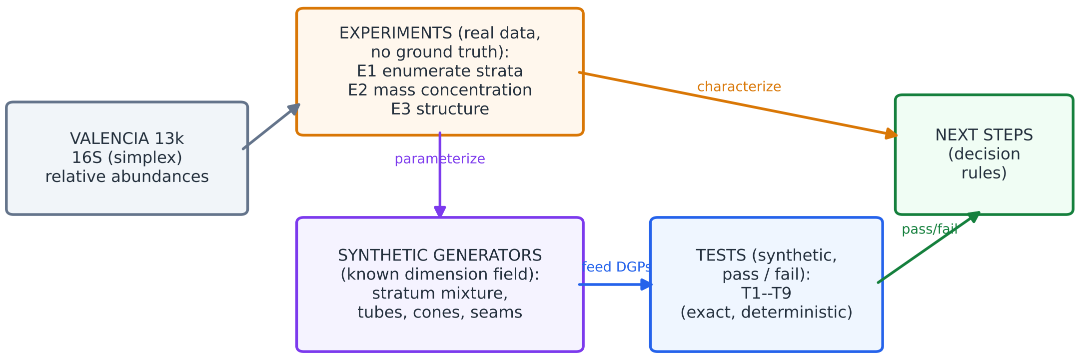

Figure 1: How the pieces fit. The real VALENCIA data is explored (experiments E1–E3, no ground truth) to characterize its stratified structure. That structure parameterizes synthetic generators with a *known* local-dimension field, which become the example set on which the deterministic tests T1–T9 run. Test outcomes and experiment findings together feed the next-step decisions.

2 The object under test, and shared conventions

We recall the estimator precisely, because the tests are written against it. Over the neighbor graph $G = (V, E)$ on n anchors, the local-dimension field $d = (d_1, \ldots, d_n) \in \{1, \ldots, D_{\max}\}^n$ minimizes

$$\widehat{d} \;=\; \arg\min_{d} \sum_{i \in V} c_i(d_i) \;+\; \lambda \sum_{(i,j) \in E} w_{ij}\, \rho(d_i, d_j), \tag{1}$$

where the per-anchor cost $c_i(d)$ is a complexity-penalized fit of dimension d to anchor i's neighborhood (e.g. the residual eigen-energy beyond d principal components measured at the smoother's bandwidth scale, plus a penalty), w_{ij} are graph weights, $\lambda \geq 0$ is the regularization strength, and the pairwise penalty is the total variation $\rho(d_i, d_j) = |d_i - d_j|$ (the ℓ_2 comparison penalty $(d_i - d_j)^2$ appears only in tests that contrast the two). Because the labels are linearly ordered integers and ρ is convex, (1) is minimized *exactly* by a single graph cut (Ishikawa). The *soft face* at scale h folds near-zero coordinates into a lower stratum, and the effective dimension is capped by the support: $d_i \leq |A_i| - 1$ where $A_i = \mathrm{supp}(x_i)$.

Conventions. The tests inherit the conventions of the experimental plan and restate only what is specific here.

- *RNG.* Every synthetic dataset is generated after an explicit seed; seeds and the package commit are stored with each result. Tests are bitwise reproducible.
- *Ground truth.* For a synthetic generator the true field is the pair (A_i, k_i) at each sample: the support pattern A_i (the naive stratum) and the designed effective dimension k_i. The "true local dimension" at scale h is k_i when h exceeds the tube thickness ρ and $|A_i| - 1$ below it (Section 5); tests fix h in the regime they intend.
- *Metrics.* Dimension misclassification rate $\frac{1}{n}\sum_i \mathbf{1}\{\widehat{d}_i \neq k_i\}$; boundary blur (defined in T2/T4); and, for the downstream test, LPS probability/continuous Truth-RMSE.
- *Coordinates.* Unless a test is specifically about the coordinate choice (T6), dimension is estimated in the cube/raw-abundance coordinate ξ, not the log-ratio u (the expository note's coordinate trap).

In plain terms Equation (1) is the whole tool in one line: each anchor proposes a dimension from its own data (c_i), neighbors are pulled toward agreement (ρ), and because dimension is a small whole number the best compromise can be found exactly. The tests below poke each part of that sentence.

3 Part A — Test specifications (synthetic, deterministic, pass/fail)

Each test follows the experimental-plan template: a falsifiable Claim; the DGP (a named synthetic generator from Section 5, whose true field (A_i, k_i) is known); the Configuration (every pinned argument); the Statistic; a numeric Acceptance criterion; and Safeguards against a confounded result. The DGP names (stratum mixture, tube, seam, cone) are defined precisely in Section 5; here we only name the regime each test needs.

3.1 T1 — the field solver returns the exact global optimum

Claim. For the total-variation penalty, the graph-cut solver of (1) returns the exact global minimum. **DGP.** Not a data generator: random small instances. Graphs with $n \leq 10$ nodes, random edges, random integer costs $c_i(d)$ for $d \in \{1, 2, 3\}$, random λ, over $R = 200$ seeds. **Configuration.** Solve by the graph-cut path; independently minimize by exhaustive enumeration of all 3^n labelings. **Statistic.** Energy gap $\Delta = E_{\text{cut}} - E_{\text{brute}}$. **Acceptance.** $\Delta = 0$ exactly on every instance (the cut is the global optimum, not an approximation). **Safeguards.** Include instances with cost ties and with λ large enough to force a single constant label and small enough to recover the unary argmin, so both extremes are exercised.

3.2 T2 — ℓ_1 keeps the boundary sharp; ℓ_2 blurs it

Claim. At a true dimension jump, the TV field transitions over a narrow band of anchors while the Laplacian (ℓ_2) field ramps over a wide one. **DGP.** The *seam* fragment: two strata of dimensions $k_1 \neq k_2$ meeting along a known boundary, with i.i.d. per-anchor estimation noise. **Configuration.** Estimate the field with $\rho = |\cdot|$ (TV) and with $\rho = (\cdot)^2$ (ℓ_2), at λ values *calibrated to equal interior denoising* (matched misclassification far from the seam), so the comparison is only about the boundary. **Statistic.** Boundary width B: the number of anchors whose fitted dimension lies strictly between k_1 and k_2 (TV labels are integers, so for TV this is the count of anchors not yet snapped to either level within a fixed corridor of the seam). **Acceptance.** $B_{\ell_1} \leq 2$ and $B_{\ell_2} \geq 5\, B_{\ell_1}$. **Safeguards.** Calibrate the two penalties to equal interior denoising first; identical noise realization for both arms (paired).

3.3 T3 — regularization reduces dimension misclassification

Claim. The TV field has lower dimension-misclassification than independent per-anchor estimation. **DGP.** The stratum-mixture generator in a near-homogeneous regime (few strata, well-separated k_A), small neighborhoods (high per-anchor variance). **Configuration.** Independent argmin of c_i vs. TV field; λ fixed (and, separately, λ chosen by T9's held-out rule). **Statistic.** Misclassification rate $\frac{1}{n}\sum_i \mathbf{1}\{\widehat{d}_i \neq k_i\}$, averaged over $R \geq 30$ replicates. **Acceptance.** TV misclassification is below independent by a predeclared relative margin (e.g. $\geq 30\%$ at the smallest neighborhood size). **Safeguards.** Same data for both arms; report the full misclassification distribution, not only the mean.

3.4 T4 — the boundary safeguard (regularization must not erase a real jump)

Claim. TV regularization denoises without smearing a true dimension jump; ℓ_2 regularization fails this and is the negative control. **DGP.** The seam fragment with a known boundary (as T2). **Configuration.** Independent, TV, and ℓ_2 fields at matched interior denoising. **Statistic.** Boundary-straddling misclassification: the error rate among anchors within graph radius r of the seam. **Acceptance.** TV increases boundary-straddling misclassification over the independent estimate by ≤ 5 percentage points, *and* ℓ_2 exceeds that bound (demonstrating the discriminator works). **Safeguards.** This test is mandatory and is the one a naive smoother fails; if ℓ_2 does not fail it, the seam is too weak — strengthen the jump.

3.5 T5 — scale-dependence and persistence

Claim. The effective dimension equals the designed k_A at scales above the tube thickness ρ and $|A| - 1$ below it; the bandwidth-tied estimate returns k_A. **DGP.** The tube fragment with known ρ and k. **Configuration.** Compute $c_i(d; h)$ over a geometric band of scales h; report the dimension-vs-scale profile and the persistence (range of h over which the chosen dimension is stable). **Statistic.** Recovered dimension-vs-scale curve compared to the designed step at ρ; the value at $h = h_{\text{band}}$. **Acceptance.** The recovered step location is within a tolerance of ρ, the high-scale plateau equals k_A, and the bandwidth-tied estimate equals k_A. **Safeguards.** ρ and k known; the test is a curve match, not a single number.

3.6 T6 — estimate dimension in ξ, not in u (the coordinate trap)

Claim. Local PCA in the cube coordinate ξ recovers the tube dimension k; local PCA in the log-ratio coordinate u inflates it near a face. **DGP.** A face-approaching tube: a stratum whose mass concentrates toward a lower face (some present coordinates near the detection limit), known k. **Configuration.** Estimate the effective rank by local PCA in ξ and, separately, in $u = \log \xi$. **Statistic.** Estimated effective rank in each coordinate. **Acceptance.** The ξ-estimate equals k within tolerance; the u-estimate is demonstrably larger ($\geq k + 1$), confirming the inflation. **Safeguards.** Known k; concentration deliberately placed near a face so the two coordinates disagree.

3.7 T7 — the support cap and the soft face

Claim. (a) The estimated dimension never exceeds the support cap; (b) the soft-face active set is monotone in scale and reduces to the hard support as $h \to 0$. **DGP.** The stratum-mixture generator with structural zeros *and* near-zeros (taxa at the detection limit). **Configuration.** Vary the soft-face scale h / threshold τ_h. **Statistic.** $\max_i \left(\widehat{d_i} - (|A_i| - 1)\right)$; and, for each sample, the effective active set $A_h(x)$ as a function of h. **Acceptance.** The cap is never violated ($\max \leq 0$); A_h is nested (monotone) in h; $A_{h \to 0} = \operatorname{supp}(x)$. **Safeguards.** Include samples sitting exactly on faces (cap tight) and samples in tubes near faces (cap loose) so both regimes are tested.

3.8 T8 — the frontier (no impossible high-dimensional islands)

Claim. The field contains no geometrically impossible isolated high-dimensional anchor (one whose neighbors are all lower-dimensional), which a proper stratification forbids by lower-semicontinuity. **DGP.** The cone fragment: a stratum whose dimension drops at a known singular point/locus (a faithful lower-semicontinuous structure). **Configuration.** The TV field, and the poset-aware (partially-ordered-label) field. **Statistic.** Number of spurious islands (a high-dimensional anchor all of whose graph neighbors have strictly lower fitted dimension, away from the true high-dimensional region). **Acceptance.** The poset-aware field has zero spurious islands; the TV field's island count is reported (it may permit a few; document the gap that motivates the poset-aware solve). **Safeguards.** The cone has a known frontier; islands are counted only away from the genuinely high-dimensional bulk.

3.9 T9 — downstream accuracy (the field is a means, not an end)

Claim. An LPS using the regularized dimension field attains lower Truth-RMSE than LPS using independent per-anchor dimension or a fixed global dimension. **DGP.** The stratum-mixture generator carrying a known response surface f (continuous and binary variants), so probability/continuous Truth-RMSE is defined. **Configuration.** Three LPS arms identical except for the chart-dimension policy: (i) independent per-anchor, (ii) fixed global, (iii) the TV field; the field's λ and scale selected by held-out Truth-RMSE. **Statistic.** Truth-RMSE on held-out points (and, for binary, probability Truth-RMSE), over $R \geq 30$ replicates, paired. **Acceptance.** The TV-field arm is below both baselines by a predeclared margin; ties or losses on near-manifold (homogeneous-dimension) generators are acceptable and expected. **Safeguards.** Match geometry, response realization, folds, support, kernel, degree, ridge across arms; vary only the dimension policy. Selecting λ by held-out predictive error (not by an intrinsic dimension score) is mandatory — the field exists to help prediction.

> **In plain terms** The tests walk up a ladder: the solver is exact (T1); the right penalty keeps edges sharp (T2, T4) while cleaning noise (T3); the estimate respects scale (T5), the right coordinate (T6), the support cap and soft face (T7), and the geometry's pecking order (T8); and, finally, all of it actually makes the smoother more accurate (T9). T9 is the one that matters in the end — the rest exist to make T9 trustworthy.

4 Part B — Experiment specifications (real VALENCIA, exploratory)

These run on the France et al. VALENCIA 16S collection ($n \approx 13{,}231$ vaginal samples), restricted to *relative abundances of phylotypes* — each sample closed to sum one, so it is a point of the simplex and nothing else (no host covariates, no counts beyond what defines presence). They have no ground truth; each produces tables, figures, and a decision rule for what to build or prioritize next. Each follows the template Objective / Procedure / Outputs / Decision.

4.1 E1 — Enumerate the naive strata

Objective. Map the face-lattice structure the data actually occupy: which support patterns (naive strata) occur, how heavily, and at what face dimensions. **Procedure.**

1. Close each sample to the simplex. Fix a presence threshold τ and define the support $A_\tau(x) = \{i : x_i > \tau\}$; *sweep* τ over a grid (e.g. 0, and relative-abundance / read-depth-aware thresholds) because "present" is not crisp at the detection limit.
2. Enumerate the distinct support patterns $\{A\}$; for each record occupancy n_A, size $|A|$, and nominal face dimension $|A| - 1$.
3. Refine: group strata by their L^∞ cell (dominant/argmax taxon), and cross-tabulate against the VALENCIA community-state-type (CST) label.

Outputs. An occupancy-ranked table of strata; the face-dimension distribution; the count of distinct strata and the occupancy concentration (how few strata hold, say, 90% of samples) as a function of τ (Figure 2); the L^∞-cell and CST cross-tabs.

Decision rule If a small number of strata hold most of the mass, focus all synthetic modeling and tests on those. If the face-dimension distribution is concentrated at low dimensions, the non-manifold structure is mild and a global chart dimension may suffice (*lowers* the priority of the whole field idea). If it is broad, local, varying dimension is warranted. The τ-sensitivity quantifies how "soft" the faces are — how much the stratum assignment moves as the threshold moves — and therefore how much the soft-face machinery (T7) is needed.

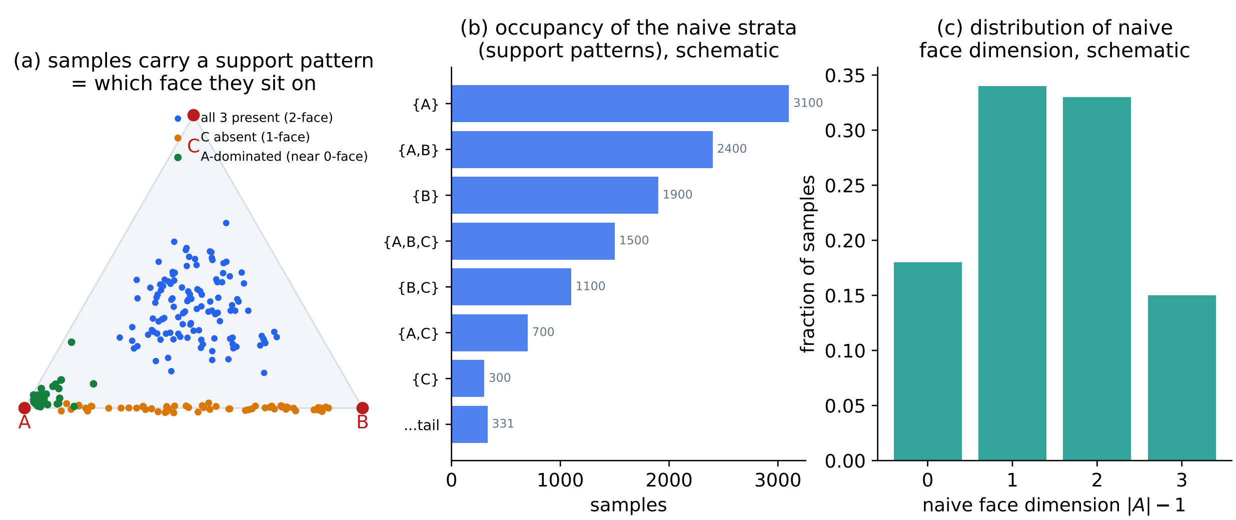

Figure 2: Schematic of the E1 outputs. (a) Each sample carries a support pattern — which phylotypes are present — equivalently which face of the simplex it sits on. (b) Occupancy of the naive strata, expected to be heavy-tailed (a few dominant patterns). (c) The distribution of nominal face dimension $|A| - 1$. (Illustrative values, not computed from the data.)

4.2 E2 — Mass concentration around lower-dimensional faces

Objective. For each well-occupied stratum, quantify how far *below* its nominal face dimension the data actually live — the gap that the local-dimension field exists to recover. **Procedure.** For each stratum A with $n_A \geq n_{\min}$, restrict to its samples and work in the cube coordinate ξ of the present parts (and, for the T6 contrast, the log-ratio u):

1. *Effective rank* k_A: the eigenvalue spectrum of the within-stratum covariance; report the scree and a gap-based k_A, alongside the nominal $|A| - 1$.
2. *Per-sample local effective dimension*: local PCA at the bandwidth scale, giving a distribution of effective dimension *within* the stratum.
3. *Near-face mass*: for each lower face $B \subset A$, the fraction of stratum-A samples within ϵ of B, i.e. with $\min_{i \in A \setminus B} \xi_i < \epsilon$; sweep ϵ.
4. *Min present-abundance law*: the distribution of $\min_{i \in A} x_i$ (how often a present taxon hugs the detection limit), and the within-stratum anisotropy (largest/smallest eigenvalue = a tube-thickness estimate ρ_A).

Outputs. Per stratum: the (nominal $|A| - 1$) vs. (effective k_A) gap (Figure 3); near-face mass curves and min-abundance histograms (Figure 4); the tube-thickness estimates ρ_A.

Decision rule The size of the (nominal − effective) gap, per stratum, is the central number. Large gaps mean the data genuinely concentrate below their face dimension, so the effective-dimension reduction is high-value and the field should be built and tested in earnest; the measured k_A and ρ_A become the parameters of the synthetic generators (E3, Section 5). Gaps near zero mean the faces are "full" and the support cap (T7) already suffices — deprioritize the within-face reduction and keep only the soft-face and the cap.

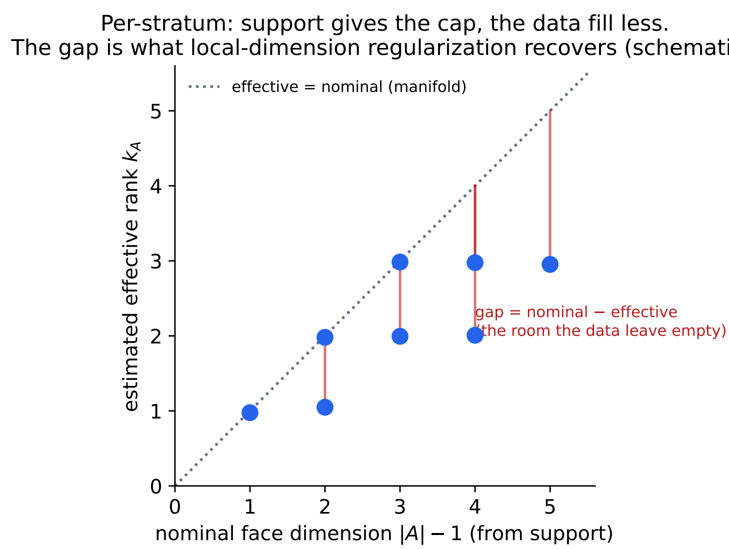

Figure 3: Schematic of the headline E2 statistic. For each stratum, the support gives a nominal dimension (the cap, on the diagonal); the data fill fewer effective directions (below the diagonal). The vertical gaps are exactly what local-dimension regularization is meant to recover. (Illustrative.)

4.3 E3 — Remaining structure for synthetic modeling

Objective. Collect the rest of the statistics needed to make the synthetic generators realistic.
Procedure. Estimate, from the data: the stratum distribution $p(A)$ (E1); per-stratum location and shape — mean μ_A and covariance Σ_A in a log-ratio chart (a logistic-normal fit), from which

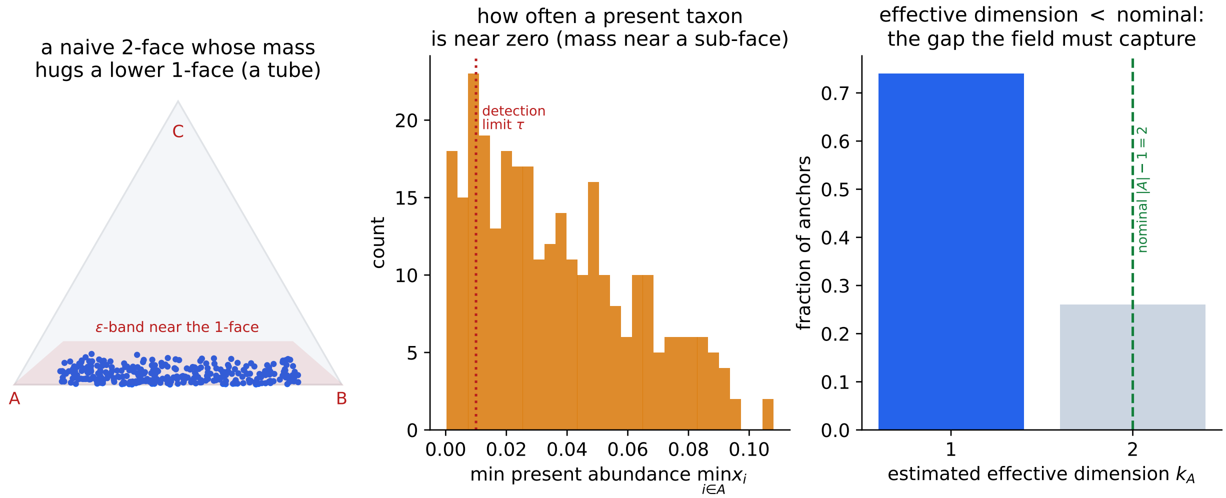

Figure 4: Schematic of the per-stratum E2 diagnostics: a naive face whose mass hugs a lower face (a tube); the distribution of the smallest present abundance (mass near a sub-face when it piles up at the detection limit); and the within-stratum effective-dimension distribution sitting below the nominal value. (Illustrative.)

the top-k_A loading directions B_A and the residual thickness ρ_A follow (E2); dominant-taxon (L^∞-cell) frequencies and the near-tie frequency (how often the top two taxa are close — the soft-cell regime); pairwise presence co-occurrence (the support-layer structure, shared with the chart-aware association design); and the same quantities conditional on CST. **Outputs.** A single parameter bundle $\{p(A), \mu_A, B_A, \rho_A, k_A,$ dominant-taxon and near-tie frequencies, co-occurrence$\}$ that fully specifies the generators of Section 5.

Decision rule Before any test built on the generators is trusted, run a posterior-predictive fidelity check: simulate from the bundle and compare synthetic vs. real marginals (per-taxon abundance laws, stratum occupancy, dominant-taxon frequencies, co-occurrence). If the synthetic data does not reproduce the real marginals, the generators — and the example set built on them — are not yet realistic, and the bundle must be refined before the test outcomes mean anything about VALENCIA.

In plain terms The experiments are a measuring exercise, not a graded one. We are taking the real microbiome data apart to answer: how many distinct “which-taxa-are-present” patterns are there, how crowded is each, and within each, do the present taxa really spread out or do they huddle near an edge? Those measurements are the recipe we then use to cook realistic fake data — and a good cook tastes the dish (the fidelity check) before serving it.

5 Part C — The VALENCIA-inspired synthetic example set

These generators expand the non-manifold example set used in previous LPS tests (replacing the toy “edge-glued-to-sheet” object with simplex-resident, VALENCIA-shaped objects). Every one carries a *known* local-dimension field, which is what makes the tests of Section 3 possible.

5.1 The stratum-mixture generator (the realistic workhorse)

DGP 5.1 (Stratum mixture)**.** Fix the parameter bundle from E3. To draw one sample:

1. draw a support pattern $A \sim p(A)$ with $|A| = m$ present parts;
2. in a log-ratio chart of the present parts (ILR, or the L^∞ max-reference cube coordinate of the companion design), $\mathbb{R}^{m-1}$, draw

$$y \;=\; \mu_A \;+\; B_A\, z \;+\; \rho_A\, \varepsilon, \qquad z \sim \mathcal{N}(0, I_{k_A}),\;\; \varepsilon \sim \mathcal{N}(0, I_{m-1}),$$

   with $B_A \in \mathbb{R}^{(m-1)\times k_A}$ of orthonormal columns (the face's k_A active tangent directions) and ρ_A small (the tube thickness). The chart covariance is $\Sigma_A = B_A B_A^\top + \rho_A^2 I$, with k_A large eigenvalues and $m - 1 - k_A$ small (ρ_A^2) ones;
3. map y back to the simplex face F_A (inverse chart), set absent parts to 0, and close to sum one.

Ground truth. The naive stratum is A; the effective local dimension at the smoother's bandwidth is k_A (we chose it). At a scale finer than ρ_A the dimension reads $m - 1$, coarser than ρ_A it reads k_A — the designed scale step that T5 checks. A response surface for T9 is attached on the latent factors, e.g. continuous $f = g(z)$ or binary $p = \mathrm{expit}(\eta(z))$.

5.2 Single-phenomenon fragments (clean unit tests)

Each fragment isolates one effect with a known field, for the corresponding test.

DGP 5.2 (Tube)**.** One stratum, $m = 3$, $k_A = 1$: a curve inside a 2-face, thickness ρ tunable. Feeds T5 (scale) and T6 (coordinate trap; place the curve heading toward a 1-face so ξ and u disagree).

DGP 5.3 (Seam)**.** Two strata of dimensions $k_1 \neq k_2$ sharing a sub-face, glued so that crossing the shared face the true dimension jumps along a *known* boundary. Feeds T2 and T4 (the boundary tests); the jump magnitude $|k_1 - k_2|$ is a knob.

DGP 5.4 (Cone)**.** A stratum whose effective dimension decreases continuously to a singular point (a 2-region pinching toward a vertex), a faithful lower-semicontinuous structure with a known frontier. Feeds T8 (the no-island / poset test).

DGP 5.5 (Near-tie ridge)**.** Two dominant taxa held near-equal so the L^∞ argmax is unstable, populating the soft-cell regime. Stresses the interaction of the dimension field with the soft atlas; an auxiliary generator for robustness checks.

In plain terms We build the fake data the way the real data is built — pick which microbes are present, then spread the sample out on that face — but with one twist we control: we decide, for each face, how many directions the data really spreads in (the "effective dimension"), and how thin the leftover directions are (the "tube"). Because we set those numbers, we know the answer key. The fragments are stripped- down versions, each built to make exactly one test fair: a single thin curve, a single seam where the dimension jumps, a single cone that pinches to a point.

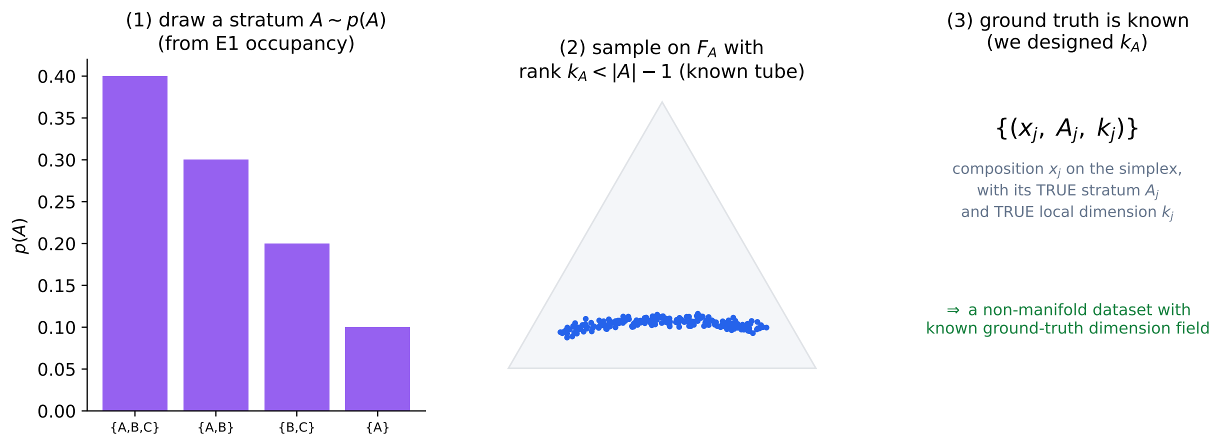

Figure 5: The stratum-mixture generator (DGP 5.1). A stratum is drawn from the E1 occupancy; a composition is sampled on that face with a covariance of controlled effective rank k_A (a known tube); the output carries its true stratum A_j and true local dimension k_j, so the dataset is non-manifold *and* gradeable. (Schematic.)

6 Part D — Sequencing and the decision map

1. **E1 first** (cheap). It alone can lower the priority of the whole program: if the real data's face-dimension structure is mild, a global chart dimension may be enough and the field is not worth building.
2. **E2 next** — the decisive measurement. The (nominal − effective) gap is the quantity that justifies, or deflates, the local-dimension field. Its k_A, ρ_A feed the generators.
3. **E3 + fidelity check.** Assemble the parameter bundle; verify the generators reproduce the real marginals before trusting any downstream test.
4. **Build the generators** (DGPs 5.1–5.5).
5. **Run the tests**, T1 first (a pure unit gate on the solver), then T2–T8 (the property tests), and T9 last (the payoff: does the field improve LPS accuracy on VALENCIA-realistic data?).

The outcomes map to next steps. If E2 shows large gaps *and* T9 shows the field helps on the realistic generator, promote `local.auto` plus the ℓ_1 field to a routine policy and proceed to a real-data LPS evaluation (where the field is judged only by held-out predictive accuracy, never by an intrinsic dimension score). If E2 shows small gaps, keep only the support cap and soft face and shelve the within-face reduction. If T8 shows the TV field producing geometrically impossible islands on the cone, invest in the poset-aware solve; otherwise TV suffices. And throughout, the synthetic example set built here becomes the permanent regression suite for the dimension field, just as the toy stratified object was for the earlier LPS tests — only now shaped like the data we actually care about.

Decision rule The single go/no-go gate for the whole effort is the conjunction *(E2 gap is large)* $\wedge$ *(T9 improves Truth-RMSE on the stratum-mixture generator)*. If either fails, the local-dimension field is not worth carrying into production for vaginal 16S, and the effort redirects to the cheaper soft-face-and-cap refinement of the chart-aware design.

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